

Mutually Independent Hamiltonian Cycles of $C_m \times C_n$ when m, n Are Odd

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Abstract

A Hamiltonian cycle of a graph G is a cycle which contains all vertices of G . Two Hamiltonian cycles $C_1 = \langle u_0, u_1, u_2, \dots, u_{n-1}, u_0 \rangle$ and $C_2 = \langle v_0, v_1, v_2, \dots, v_{n-1}, v_0 \rangle$ in G are independent if $u_0 = v_0, u_i \neq v_i$ for all $1 \leq i \leq n-1$. If any two Hamiltonian cycles of a Hamiltonian cycles set $C = \{C_1, C_2, \dots, C_k\}$ are independent, we call C is mutually independent. The mutually independent Hamiltonicity $IHC(G) = k$ means there exists a maximum integer k such that there exists k -mutually independent Hamiltonian cycles start from any vertex of G . In this paper, we prove that $IHC(C_m \times C_n) = 4$, for $m, n \geq 3$, m, n are odd.

1 Introduction

Cycle is the fundamental class of network topologies. When we design an algorithm for parallel and distributed computing, cycle is a suitable network topology [8, 17, 18]. It has many efficient parallel algorithms designed on cycle for solving various algebraic problems, graph problems, and some parallel applications, such as those in image and signal processing, which can be found in [1, 8]. If a network can be embedded on cycle, then we can use some algorithms as control/data flow structures for distributed computing in the network. By this transformation, these algorithms which designed on cycle can be executed on the network.

When designing a structure in interconnection network, we often transform this problem to graph for discussing and studying. Formally, a processor will be transformed to be a vertex and the connection between two processors will be transformed to be an edge. By this transformation, it can transform an interconnection network to a graph.

For definitions and notations, we follow [3]. A graph $G = (V, E)$, V, E are two finite sets, where V is called vertex set and E is called edge set, which defined as $E \subseteq \{(u, v) : (u, v) \text{ is an unordered pair of } V\}$. Two vertices u, v are adjacent if $(u, v) \in E(G)$, $u, v \in V(G)$. A path $P[v_0, v_k] = \langle v_0, v_1, \dots, v_k \rangle$ in G , define as a sequence of adjacent vertices and for all $0 \leq i < j \leq k$, $v_i \neq v_j$. Let $P(i) = v_{i-1}$ be the i th vertex of path P . A cycle $C[v_0, v_k] = \langle v_0, v_1, \dots, v_k \rangle$ for $k \geq 3$, is a path expect $v_0 = v_k$. In G , a cycle is Hamiltonian cycle if it contains all vertices of G . Two Hamiltonian cycles $C_1 = \langle u_0, u_1, u_2, \dots, u_n, u_0 \rangle$ and $C_2 = \langle v_0, v_1, v_2, \dots, v_n, v_0 \rangle$ in G are independent if $u_0 = v_0, u_i \neq v_i$ for all $1 \leq i \leq n$. If any two Hamiltonian cycles of a Hamiltonian cycles set $C = \{C_1, C_2, \dots, C_k\}$ are independent, we call C is mutually independent. The mutually independent Hamiltonicity $IHC(G) = k$ of a graph G means there exists a maximum integer k such that there exist k -mutually independent Hamiltonian cycles start from any vertex of G . In this paper, we define C_n is a cycle with $|V(C_n)| = n$.

Broadcasting is an information dissemination process that involves one node in a network sending pieces of information to all other nodes in the network. Constructing a set of k -mutually independent Hamiltonian cycles enables us to efficiently broadcast a message formed of k pieces on a graph G . The common starting vertex acts as the source to send the k pieces of the given message along the k parallel Hamiltonian cycles. This broadcasting can be done in $O(n)$ time using an all-port model.

Given two graph $G = (V_G, E_G), H = (V_H, E_H)$, the Cartesian product of G and H , $G \times H$, is a graph which $V = V_G \times V_H$, $E = \{((p_1, q_1), (p_2, q_2)) : (p_1, q_1), (p_2, q_2) \in V, \text{ and } (p_1 = p_2, (q_1, q_2) \in E_H, \text{ or } q_1 = q_2, (p_1, p_2) \in E_G)\}$. For a vertex $u \in V(G)$, the neighbor of u , $N_G(u)$, is the set $\{v : (u, v) \in E(G)\}$. And $\deg_G(u) = |N_G(u)|$.

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is called the *degree* of u . In a graph G , we use $\Delta(G)$ and $\delta(G)$ to denote the *maximum degree* and *minimum degree* of G such that $\Delta(G) = \text{maximum degree}$, $\delta(G) = \text{minimum degree}$. When we consider $\text{IHC}(G)$, these mutually independent Hamiltonian cycles are start from any vertex of G , then we can get a property as follows:

Remark 1. For any graph G , $\text{IHC}(G) \leq \delta(G)$.

Mutually independent Hamiltonian cycle problem is an important property for graph theory, there are many studies for this problem on different graph [2, 5–7, 9, 10, 12, 13]. For hypercube, it has been studied in [4, 14]. For the pancake graph and the star graph, there is a discussing on it [11]. In 2011, we study this problem on $C_n \times C_n$ for n is positive integer [16]. In this paper, we study on mutually independent Hamiltonicity of $C_m \times C_n$, and we prove that $\text{IHC}(C_m \times C_n) = 4$, for $m, n \geq 3$ are odd.

2 Symbol Definition

Let $Z_n = \{0, 1, 2, \dots, n-1\}$, $n \in \mathbb{N}$ and let $V(C_n) = Z_n$. We refer to *toroidal mesh graph* [15] and define graph $C_m \times C_n$, for $m, n \geq 3$ as follows: The vertex set $V(C_m \times C_n) = \{(x, y) : x \in Z_m, y \in Z_n\}$; The edge set $E(C_m \times C_n) = \{((u_1, u_2), (v_1, v_2)) : (u_1, u_2), (v_1, v_2) \in V(C_m \times C_n) \text{ and } u_1 = v_1, |u_2 - v_2| \in \{1, n-1\} \text{ or } u_2 = v_2, |u_1 - v_1| \in \{1, m-1\}\}$. We use this symbol definition to construct $C_6 \times C_5$ as Figure 1. A graph G is a k -regular graph if for all $x \in V(G)$, $\deg_G(x) = k$. And a graph is *vertex transitive* if every vertex can be mapped to any other vertex by some automorphism [17]. By definition, $C_m \times C_n$ is a 4-regular graph with mn vertices and it is vertex transitive.

In $C_m \times C_n$, we use C^i to denote the i th subgraph of $C_m \times C_n$ for $i \in Z_m$, where the vertex set $V(C^i) = \{(i, v) : v \in Z_n\}$; The edge set $E(C^i) = \{((i, v_1), (i, v_2)) : |v_1 - v_2| \in \{1, n-1\} \text{ and } v_1, v_2 \in Z_n\}$. Then $C_m \times C_n$ can be decomposed into m subgraph C^i for $i \in Z_m$, and each C^i is isomorphic to C_n . Let (i, x) and $(i, y) \in V(C^i)$, then we define $P_{i,x,y,+} = \langle (i, x), (i, x+1), (i, x+2), \dots, (i, y) \rangle$ is a path of C^i which is from (i, x) to (i, y) and the index of vertices are increased. And $P_{i,x,y,-} = \langle (i, x), (i, x-1), (i, x-2), \dots, (i, y) \rangle$ is a path of C^i which is from (i, x) to (i, y) and the index of vertices are decreased. Note that the operation $+$ or $-$ should be calculate on Z_n . In a

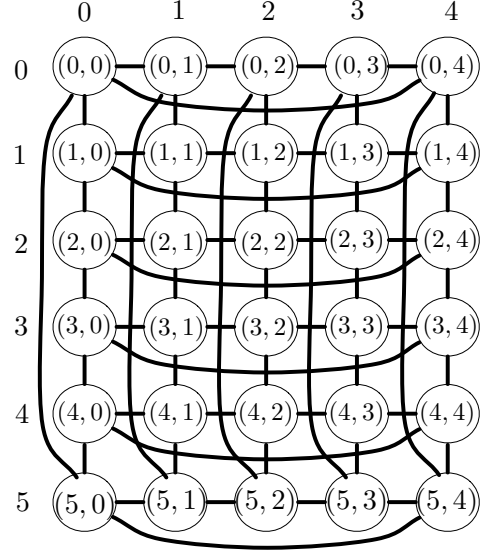


Figure 1: Symbol definition of $C_6 \times C_5$.

cycle C , we use $C(t) = x$ for $1 \leq t \leq |C|$ to denote the t th vertex of C from the start vertex.

3 Main Result

We study on the mutually independent Hamiltonicity of $C_m \times C_n$ in this section.

Lemma 1. [16] For any odd positive integer $n \geq 3$, $\text{IHC}(C_n) = 2$.

Theorem 1. [16] For any odd positive integer $n \geq 3$, $\text{IHC}(C_n \times C_n) = 4$.

Theorem 2. For $m, n \geq 3$ and m, n are odd, $\text{IHC}(C_m \times C_n) = 4$.

Proof. For $m = n$, by Theorem 1, it is proved. We consider about $m \neq n$ and without loss of generality we assume that $m > n$. Since $C_m \times C_n$ is 4-regular graph, $\text{IHC}(C_m \times C_n) \leq 4$ by Remark 1. Next, we will show that $\text{IHC}(C_m \times C_n) \geq 4$ by constructing 4-mutually independent Hamiltonian cycles from any vertex of $C_m \times C_n$. Without loss of generality, we may assume that start vertex $e = (0, 0)$ because $C_m \times C_n$ is vertex transitive. We define following paths and use them to construct 4 Hamiltonian cycles start from e .

$$\begin{aligned} T_{1,0} &= \langle (0, 0), P_{0,0,n-1,+}, (0, n-1), (1, n-1), \\ &P_{1,n-1,n-2,+}, (1, n-2), (2, n-2), P_{2,n-2,n-3,+}, \\ &(2, n-3), \dots, (n-1, 1), P_{n-1,1,0,+}, (n-1, 0) \rangle, \\ T_{1,1} &= \langle (m-2, 0), P_{m-2,0,1,-}, (m-2, 1), (m-1, 1), \\ &P_{m-1,1,0,+}, (m-1, 0), (0, 0) \rangle, \end{aligned}$$

$T_{1,2} = \langle (m-4, 0), P_{m-4,0,1,-}, (m-4, 1), (m-3, 1), P_{m-3,1,2,-}, (m-3, 2), (m-2, 2), P_{m-2,2,1,+}, (m-2, 1), (m-1, 1), P_{m-1,1,0,+}, (m-1, 0), (0, 0) \rangle$,

$T_{1,3} = \langle (m-6, 0), P_{m-6,0,1,-}, (m-6, 1), (m-5, 1), P_{m-5,1,2,-}, (m-5, 2), (m-4, 2), P_{m-4,2,3,-}, (m-4, 3), (m-3, 3), P_{m-3,3,2,+}, (m-3, 2), (m-2, 2), P_{m-2,2,1,+}, (m-2, 1), (m-1, 1), P_{m-1,1,0,+}, (m-1, 0), (0, 0) \rangle$,

$T_{1,4} = \langle (n, 0), P_{n,0,1,-}, (n, 1), (n+1, 1), P_{n+1,1,0,+}, (n+1, 0), (n+2, 0), P_{n+2,0,1,-}, (n+2, 1), \dots, (m-7, 1), P_{m-7,1,0,+}, (m-7, 0) \rangle$,

$T_{2,0} = \langle (0, 0), (1, 0), P_{1,0,1,-}, (1, 1), (2, 1), P_{2,1,2,-}, (2, 2), (3, 2), P_{3,2,3,-}, (3, 3), \dots, (n-1, n-2), P_{n-1,n-2,n-1,-}, (n-1, n-1), (n, n-1), P_{n,n-1,n-2,+}, (n, n-2) \rangle$,

$T_{2,1} = \langle (n+1, n-2), P_{n+1,n-2,n-1,-}, (n+1, n-1), (n+2, n-1), P_{n+2,n-1,n-2,+}, (n+2, n-2), (n+3, n-2), P_{n+3,n-2,n-1,-}, (n+3, n-1), (n+4, n-1), P_{n+4,n-1,n-2,+}, (n+4, n-2), \dots, (m-1, n-2), P_{m-1,n-2,n-1,-}, (m-1, n-1), (0, n-1), P_{0,n-1,0,-}, (0, 0) \rangle$,

$T_{3,0} = \langle (0, 0), (m-1, 0), P_{m-1,0,n-1,+}, (m-1, n-1), (m-2, n-1), P_{m-2,n-1,n-2,+}, (m-2, n-2), (m-3, n-2), P_{m-3,n-2,n-3,+}, (m-3, n-3), \dots, (m-n, 1), P_{m-n,1,0,+}, (m-n, 0) \rangle$,

$T_{3,1} = \langle (m-n-1, 0), P_{m-n-1,0,1,-}, (m-n-1, 1), (m-n-2, 1), P_{m-n-2,1,0,+}, (m-n-2, 0), (m-n-3, 0), P_{m-n-3,0,1,-}, (m-n-3, 1), \dots, (2, 1), P_{2,1,0,+}, (2, 0), (1, 0), P_{1,0,1,-}, (1, 1), (0, 1), P_{0,1,0,+}, (0, 0) \rangle$,

$T_{4,0} = \langle (0, 0), P_{0,0,1,-}, (0, 1), (m-1, 1), P_{m-1,1,2,-}, (m-1, 2), (m-2, 2), P_{m-2,2,3,-}, (m-2, 3), \dots, (m-n+2, n-2), P_{m-n+2,n-2,n-1,-}, (m-n+2, n-1), (m-n+1, n-1), P_{m-n+1,n-1,n-2,+}, (m-n+1, n-2) \rangle$,

$T_{4,1} = \langle (m-n, n-2), P_{m-n,n-2,n-1,-}, (m-n, n-1), (m-n-1, n-1), P_{m-n-1,n-1,n-2,+}, (m-n-1, n-2), (m-n-2, n-2), P_{m-n-2,n-2,n-1,-}, (m-n-2, n-1), \dots, (2, n-2), P_{2,n-2,n-1,-}, (2, n-1), (1, n-1), P_{1,n-1,0,-}, (1, 0), (0, 0) \rangle$,

We construct 4-mutually independent Hamiltonian cycles H_1, H_2, H_3 and H_4 start from e as follows:

$H_2 = \langle (0, 0), T_{2,0}, (n, n-2), (n+1, n-2), T_{2,1}, (0, 0) \rangle$, $H_3 = \langle (0, 0), T_{3,0}, (m-n, 0), (m-n-1, 0), T_{3,1}, (0, 0) \rangle$, $H_4 = \langle (0, 0), T_{4,0}, (m-n+1, n-2), (m-n, n-2), T_{4,1}, (0, 0) \rangle$.

For H_1 : If $m-n=2$, $H_1 = \langle (0, 0), T_{1,0}, (n-1, 0), (m-2, 0), T_{1,1}, (0, 0) \rangle$, since $m-2=n=n-1+1$. If $m-n=4$, $H_1 = \langle (0, 0), T_{1,0}, (n-1, 0), (m-4, 0), T_{1,2}, (0, 0) \rangle$, since $m-4=n=n-1+1$. If $m-n=6$, $H_1 = \langle (0, 0), T_{1,0}, (n-1, 0), (m-6, 0), T_{1,3}, (0, 0) \rangle$, since $m-6=n=n-1+1$. If $m-n \geq 8$, $H_1 = \langle (0, 0), T_{1,0}, (n-1, 0), (n, 0) \rangle$,

$T_{1,4}, (m-7, 0), (m-6, 0), T_{1,3}, (0, 0) \rangle$. According to $H_i(t)$, $2 \leq t \leq mn$, $1 \in \{1, 2, 3, 4\}$ belong to the same subgraph C^j for some $j \in Z_n$ or not, we have following cases. Figure 2 is the illustration for these case on $C_9 \times C_5$.

Case 1 : $2 \leq t \leq n$ or $n(m-1)+2 \leq t \leq mn$.

For any $2 \leq t_1 \leq n$ and $n(m-1)+2 \leq t_2 \leq mn$. By the construction of H_1, H_2, H_3 and H_4 , we can know that $H_1(t_1), H_4(t_1) \in V(C^0)$; $H_2(t_1) \in V(C^1)$; $H_3(t_1) \in V(C^{m-1})$ and $H_1(t_2) \in V(C^{m-1})$; $H_2(t_2), H_3(t_2) \in V(C^0)$; $H_4(t_2) \in V(C^1)$. By Lemma 1, $H_1(t_1) \neq H_4(t_1)$ and $H_2(t_2) \neq H_3(t_2)$. In the same time, $H_2(t_1), H_3(t_1); H_1(t_2), H_4(t_2)$ are in different subgraph, so $H_i(t) \neq H_j(t)$, for $1 \leq i \neq j \leq 4$, and $2 \leq t \leq n$ or $n(m-1)+2 \leq t \leq mn$. That is, H_1, H_2, H_3 and H_4 are mutually independent for this case.

Case 2 : $t = in+1$, for all $1 \leq i \leq m-1$.

From the construction of 4 Hamiltonian cycles, we can know that for all $1 \leq i \leq n-1, t = in+1$, $H_1(t) = (i, n-i); H_2(t) = (i, i); H_3(t) = (m-i, n-i); H_4(t) = (m-i, i)$. Since m, n are odd, $n-i \neq i$. It is mean that when $t = in+1$ for $1 \leq i \leq n-1$ $H_1(t), H_2(t), H_3(t), H_4(t)$ are distinct.

For all $n \leq i \leq m-1, t = in+1$, $H_1(t), H_2(t) \in V(C^i)$ $H_3(t), H_4(t) \in V(C^{m-i})$. Since H_1 and H_2 are in different subgraph, we have following cases.

Case 2.1 : H_1, H_2 .

From the construction of H_1 , we can know that $H_1(t)$ is distinct in $m-n \leq 6$ and $m-n \geq 8$ for $t = in+1$ and $n \leq i \leq m-1$. Then we have following subcases.

Case 2.1.1 : $m-n \leq 6$.

For $m-n=6$ and $n \leq i \leq m-1, t = in+1$: $H_1(n^2+1) = (n, 0)$, $H_1((n+1)n+1) = (n+1, 1)$, $H_1((n+2)n+1) = (n+2, 2)$, $H_1((n+3)n+1) = (n+3, 3)$, $H_1((n+4)n+1) = (n+4, 2)$, $H_1((n+5)n+1) = (n+5, 1)$. Besides, if $i \bmod 2 = 0$, $H_2(in+1) = (i, n-1)$ and if $i \bmod 2 = 1$, $H_2(in+1) = (i, n-2)$. It is trivial that $0 \neq n-2$, $1 \neq n-1$, $2 \neq n-2$, $3 \neq n-1$, for $n \geq 3$ and n is odd. For $m-n=4$ or 2 , the proof are similar to above discussion. Hence, H_1 and H_2 are independent in this subcase.

Case 2.1.2 : $m-n \geq 8$.

For all $n \leq i \leq m-7$ and $t = in+1$: if $i \bmod 2 = 0$, $H_1(t) = (i, 1)$ and if $i \bmod 2 = 1$, $H_1(in+1) = (i, 0)$. And for $m-6 \leq i \leq m-1$: $H_1((m-6)n+1) = (m-6, 0)$, $H_1((m-5)n+1) = (m-5, 1)$, $H_1((m-4)n+1) = (m-4, 2)$, $H_1((m-3)n+1) = (m-3, 3)$, $H_1((m-2)n+1) = (m-2, 2)$, $H_1((m-1)n+1) = (m-1, 1)$. In the other hand, for all $n \leq i \leq m-1$ and $t = in+1$

: if $i \bmod 2 = 0$, $H_2(in + 1) = (i, n - 1)$ and if $i \bmod 2 = 1$, $H_2(in + 1) = (i, n - 2)$. It is trivial that $0 \neq n - 2$, $1 \neq n - 1$, $2 \neq n - 2$, $3 \neq n - 1$, for $n \geq 3$ and n is odd. Hence, H_1 and H_2 are independent in this subcase.

Case 2.2 : H_3, H_4 .

For all $n \leq i \leq m - 1$, $t = in + 1$, if $i \bmod 2 = 0$ $H_3(t) = (m - i, 1)$, $H_4(t) = (m - i, n - 1)$ and if $i \bmod 2 = 1$ $H_3(t) = (m - i, 0)$, $H_4(t) = (m - i, n - 2)$. It is easy to see that $1 \neq n - 1$ and $0 \neq n - 2$ for n is odd, then $H_3(t), H_4(t)$ are mutually independent in this subcase.

Case 3 : $in + 2 \leq t \leq in + n$, for all $1 \leq i \leq m - 2$ and $i \neq (m - 1)/2$.

For all t in this csae $H_1(t) \in V(C^i)$, $H_2(t) \in V(C^{i+1})$, $H_3(t) \in V(C^{m-i-1})$, and $H_4(t) \in V(C^{m-i})$. It is trivial that $i \neq i + 1$, $m - i - 1 \neq m - i$. Since $i \neq (m - 1)/2$, then $i \neq m - i - 1$; $i + 1 \neq m - i$. Besides, $i \neq m - i$ and $i + 1 \neq m - i - 1$, for m, n are odd. Hence, H_1, H_2, H_3 and H_4 are mutually independent in this range.

Case 4 : $in + 2 \leq t \leq (i + 1)n$, for $i = (m - 1)/2$.

When $t = ((m - 1)/2)n + 2$, $H_1(t - 1) = H_3(t)$; $H_4(t - 1) = H_2(t)$, and for all $in + 2 \leq t \leq (i + 1)n$, $i = (m - 1)/2$, $H_1(t), H_3(t) \in V(C^i)$; $H_2(t), H_4(t) \in V(C^{m-i})$. For $m - n < n - 1$, $\langle H_1(in + 1), H_1(in + 2), \dots, H_1(in + n) \rangle = \langle H_3(in + 2), H_3(in + 3), \dots, H_3(in + n + 1) \rangle = P_{i, n-i, n-i-1, +}$ and $\langle H_2(in + 2), H_2(in + 3), \dots, H_2(in + n + 1) \rangle = \langle H_4(in + 1), H_4(in + 2), \dots, H_4(in + n) \rangle = P_{m-i, i, i+1, -}$.

For $m - n \geq n - 1$, if $(m - n - (n - 1)) \bmod 4 = 0$, then $\langle H_1(in + 1), H_1(in + 2), \dots, H_1(in + n) \rangle = \langle H_3(in + 2), H_3(in + 3), \dots, H_3(in + n + 1) \rangle = P_{i, 0, 1, -}$ and $\langle H_2(in + 2), H_2(in + 3), \dots, H_2(in + n + 1) \rangle = \langle H_4(in + 1), H_4(in + 2), \dots, H_4(in + n) \rangle = P_{m-i, n-2, n-1, -}$. And if $(m - n - (n - 1)) \bmod 4 = 2$, then $\langle H_1(in + 1), H_1(in + 2), \dots, H_1(in + n) \rangle = \langle H_3(in + 2), H_3(in + 3), \dots, H_3(in + n + 1) \rangle = P_{i, 1, 0, +}$ and $\langle H_2(in + 2), H_2(in + 3), \dots, H_2(in + n + 1) \rangle = \langle H_4(in + 1), H_4(in + 2), \dots, H_4(in + n) \rangle = P_{m-i, n-1, n-2, +}$.

From the above discussion H_1, H_3 and H_2, H_4 are independent. Besides, H_1, H_3 and H_2, H_4 are in different subgraphs. Thus, H_1, H_2, H_3 and H_4 are mutually independent in this range.

From the above 4 cases, we can know that H_1, H_2, H_3 and H_4 start from e are mutually independent and $\text{IHC}(C_m \times C_n) \geq 4$ because $C_m \times C_n$ is vertex transitive. Hence, $\text{IHC}(C_m \times C_n) = 4$ can be concluded. $\square$

We construct 4-mutually independent Hamilto-

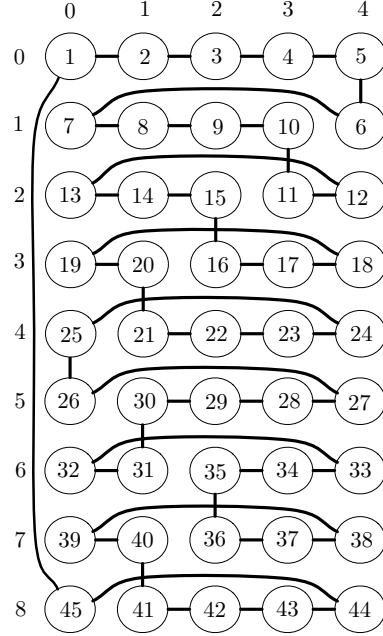


Figure 3: Hamiltonian cycles H_1 in $C_9 \times C_5$.

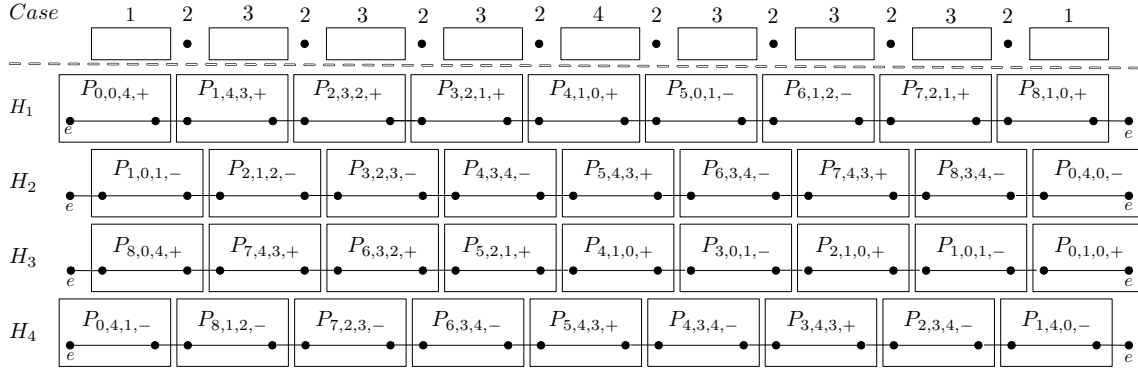
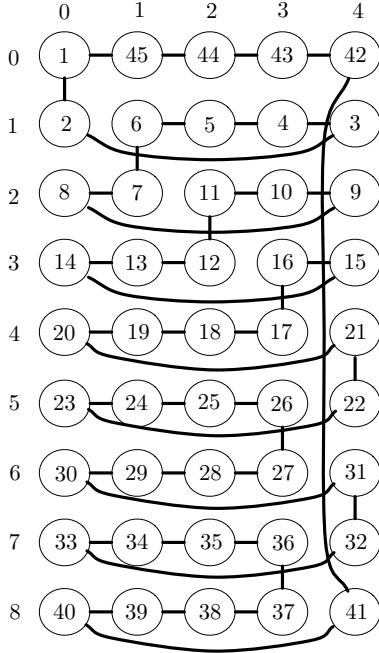
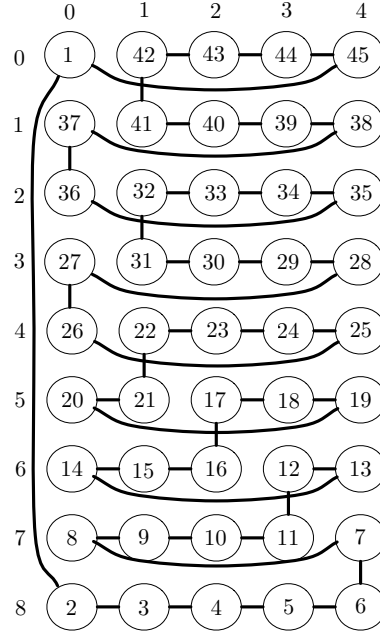
nian cycles for $C_9 \times C_5$ in Figures 3, 4, 5, 6 as an example of Theorem 1.

4 Conclusion

In this paper, we discuss the mutually independent Hamiltonian cycles of $C_m \times C_n$, for m, n are odd ≥ 3 , and get the optimal result on it. Note that a Hamilton cycles in any graph G is a cycle C_n with $n = |V(G)|$. So, if two graphs G_1 and G_2 are Hamiltonian with $|V(G_1)| = m$, $|V(G_2)| = n$ both are odd and ≥ 3 , then graph $C_m \times C_n$ will be a spanning subgraph of $G_1 \times G_2$. Hence, we have the following corollary.

Corollary 1. For any graphs G_1, G_2 , if G_1 and G_2 are Hamiltonian and $|V(G_1)|, |V(G_2)|$ are odd and ≥ 3 , then $\text{IHC}(G_1 \times G_2) \geq 4$.

In the future, discussing the mutually independent Hamiltonian cycles of $C_m \times C_n$, for $m, n \geq 4$ and m, n are even is an interesting question. Furthermore, we can discuss $\text{IHC}(G)$ for $G = C_n^k = C_n \times C_n \times \dots \times C_n$, or more general graph $C_{n_1} \times C_{n_2} \times \dots \times C_{n_k}$, for any positive integer $n_1 \geq n_2 \geq \dots \geq n_k \geq 3$.


 Figure 2: The construction of 4 Hamiltonian cycles of $C_9 \times C_5$ start at e

 Figure 4: Hamiltonian cycles H_2 in $C_9 \times C_5$.

 Figure 5: Hamiltonian cycles H_3 in $C_9 \times C_5$.

5 Acknowledgment

This research was supported in part by the National Science Council of the Republic of China under grant NSC 100-2221-E-260-024-.

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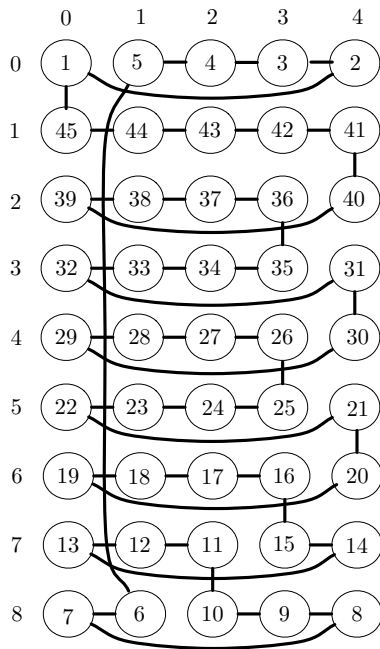


Figure 6: Hamiltonian cycles H_4 in $C_9 \times C_5$.

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